TrES-3b

R-0471

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_230908` · created 2026-07-20T01:52:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460195.6712 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.79 +/- 2.21 [%]
Orbital Inclination (inc)	82.66 +/- 2.82
Airmass coefficient 1 (a1)	2.18 +/- 0.55
Airmass coefficient 2 (a2)	-0.74 +/- 0.19
Scatter in the residuals of the lightcurve fit is	22.69 %
Best Comparison Star	#1 - [367, 347]
Optimal Aperture	4.81
Optimal Annulus	18.05
Transit Duration (day)	0.059 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.059 d vs 0.0567 d (deviation 0.09σ)
Mid-time O–C	-5.2 min
Depth SNR	3.1
Residual scatter	22.69 %

Light curve

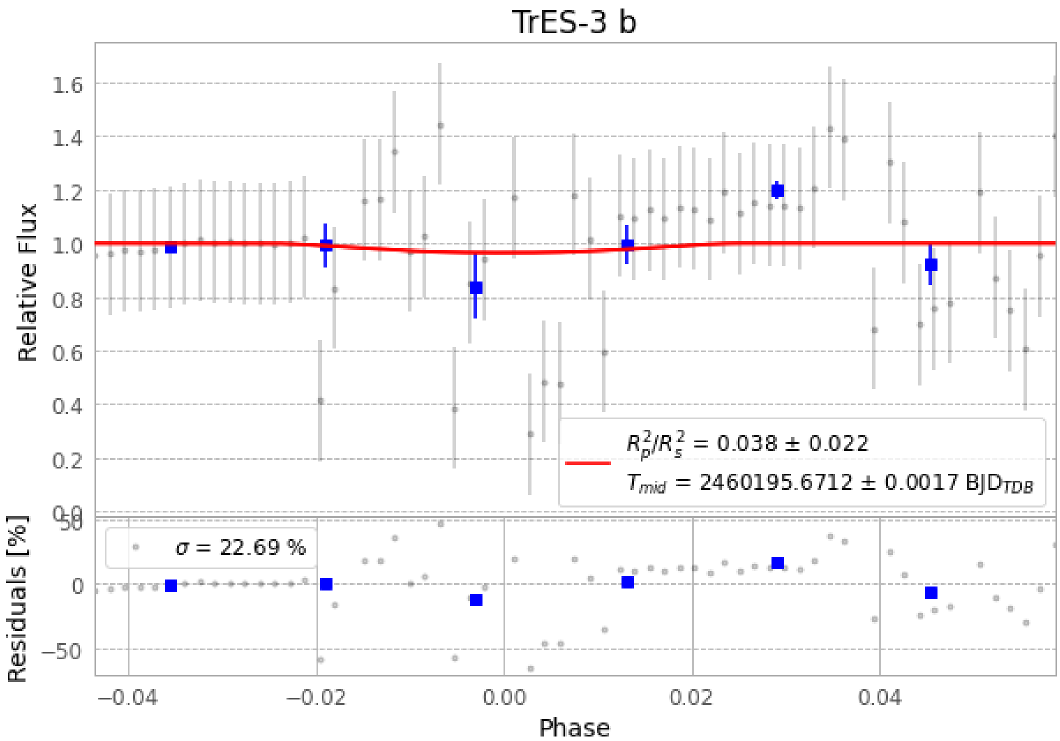

Figure 1 — FinalLightCurve_TrES-3 b_2023-09-07.png (SHA-256 aa343bb4259868db…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [257, 191], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ebe042207f5ead10…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

65 FITS frames (43 MB), `TRES-3230908024215.FITS` → `TRES-3230908055415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230908.log (d8e84ea6a83c…), FinalLightCurve_TrES-3 b_2023-09-07.png (aa343bb42598…), FinalLightCurve_TrES-3 b_2023-09-07.csv (a31d7e71c56e…)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)